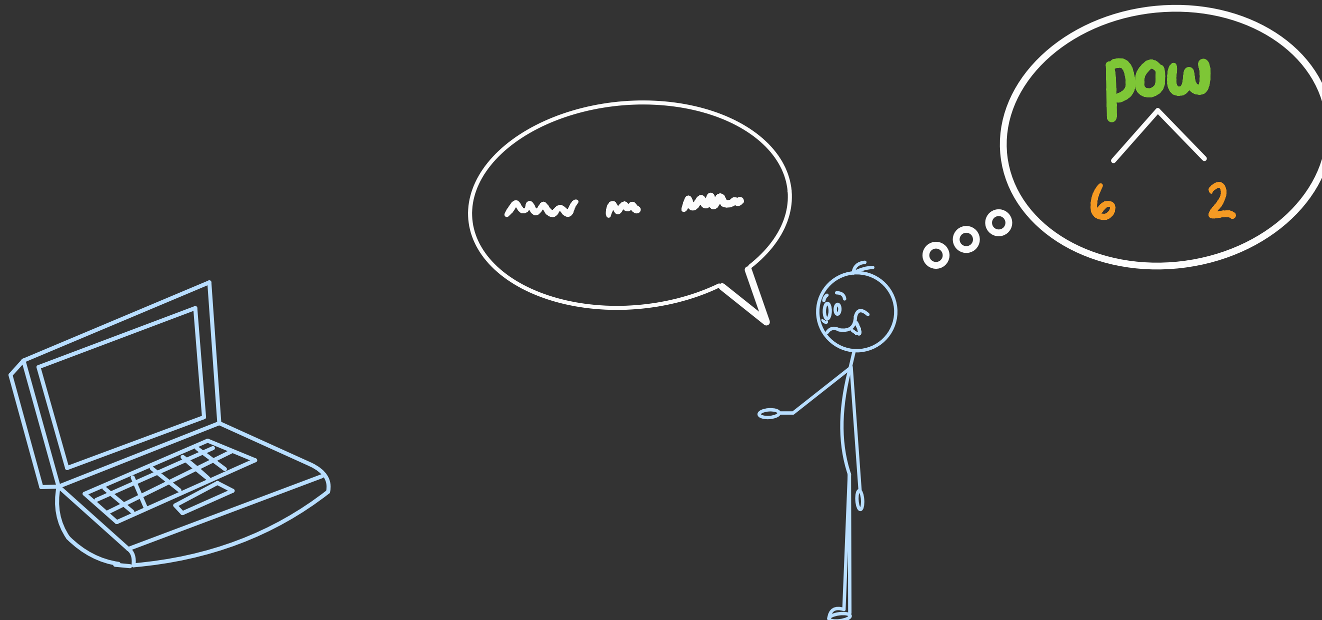
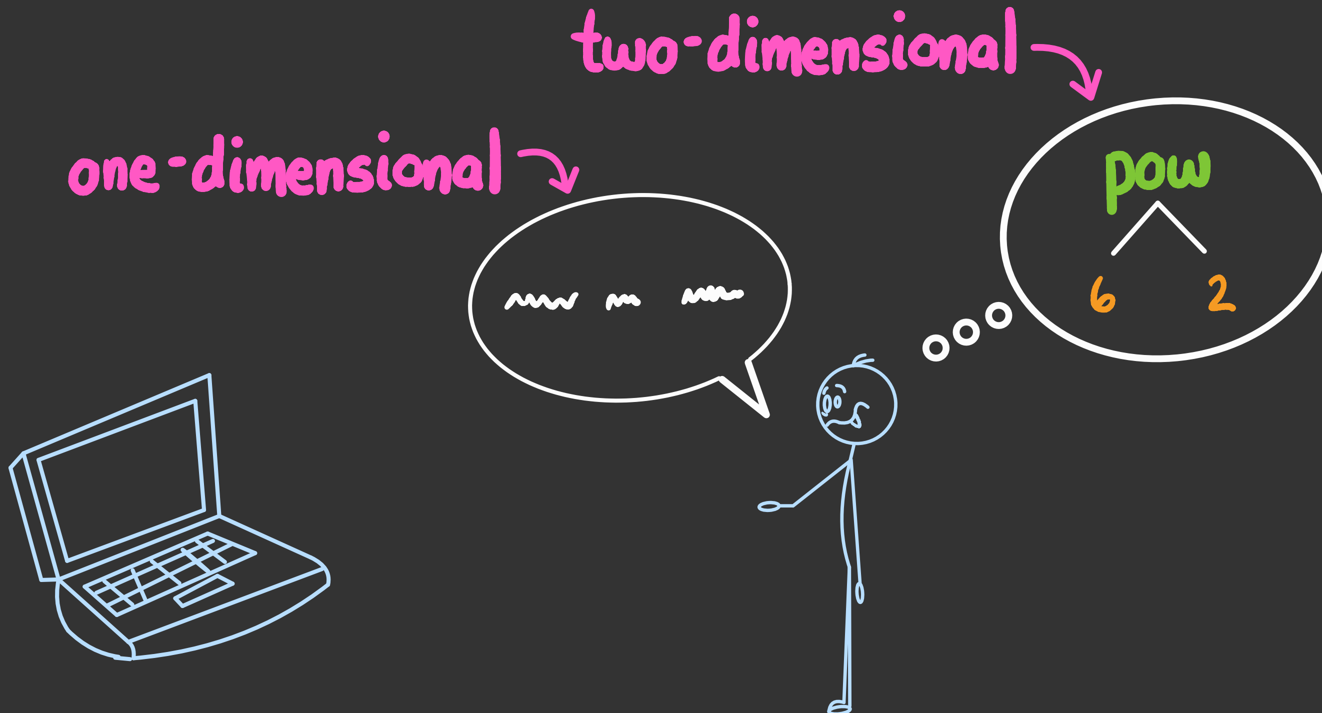
Flattening expressions

CSCI 134



how do we communicate an expression tree

to a computer?



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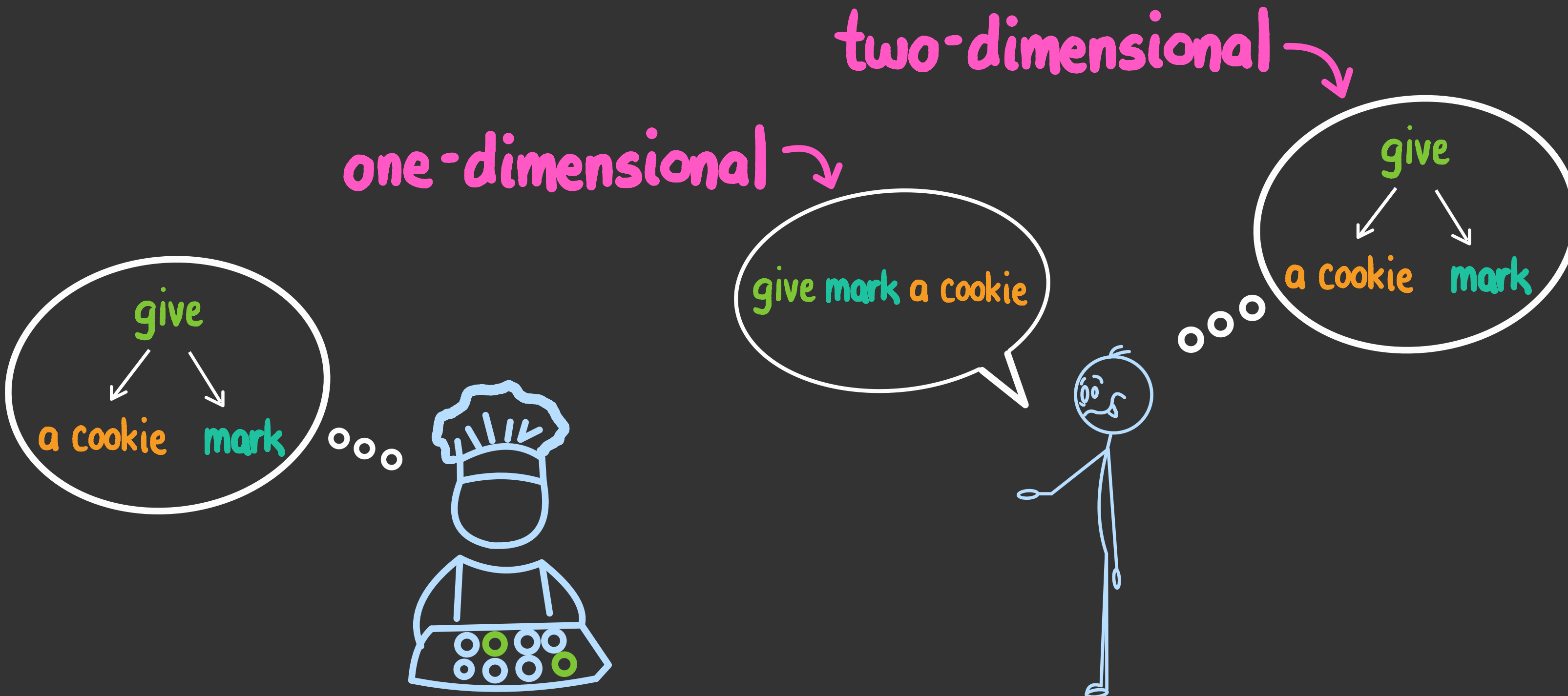


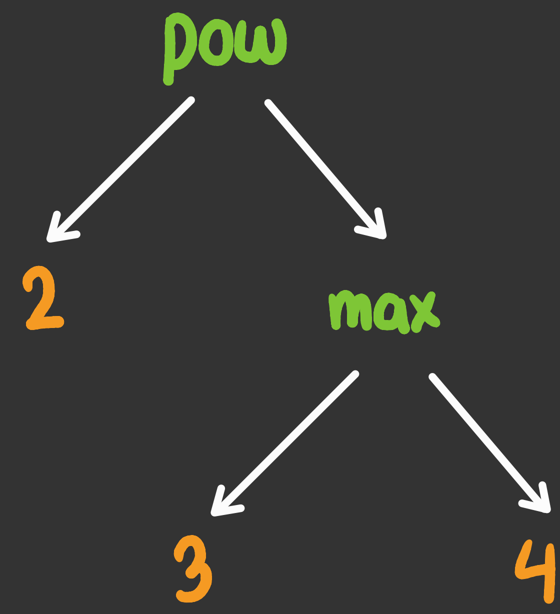
in english, we agree on a particular word order:
the verb, then the recipient, then the object

but this convention can result in

ambiguity and **confusion**

sidebar:

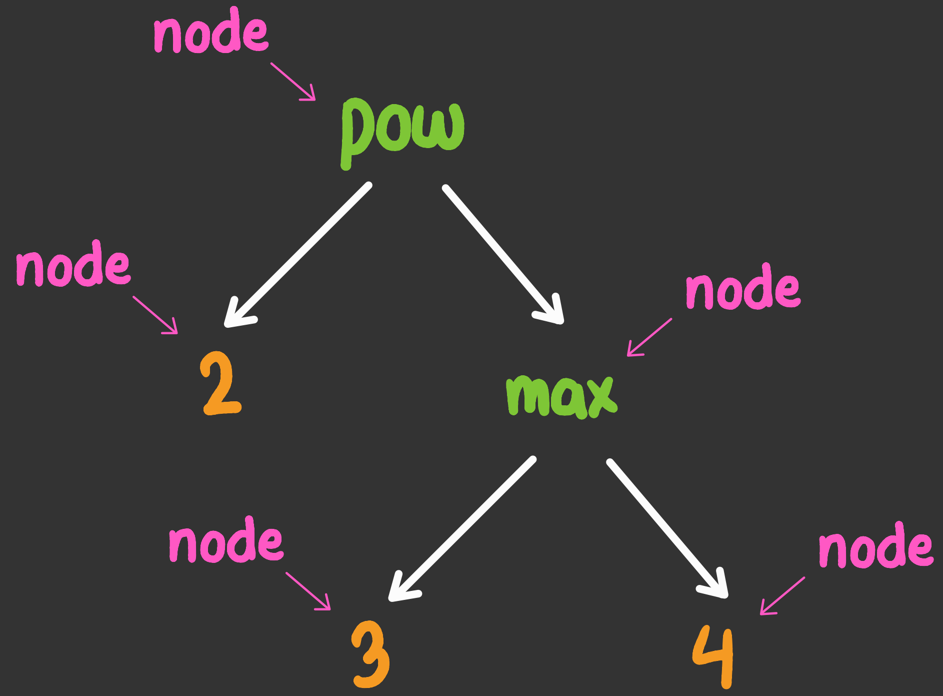
how to talk about trees



sidebar:

how to talk about trees

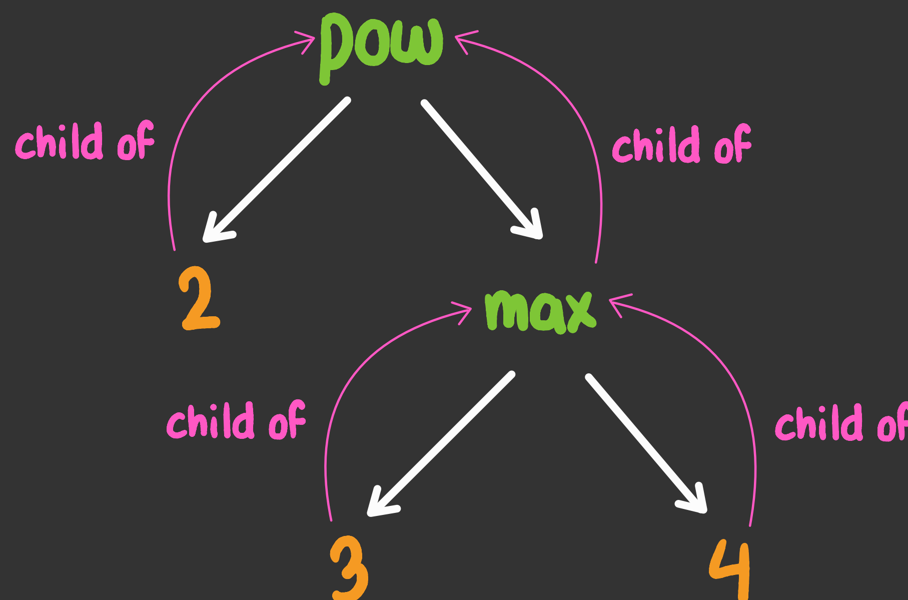
nodes



sidebar:

how to talk about trees

children

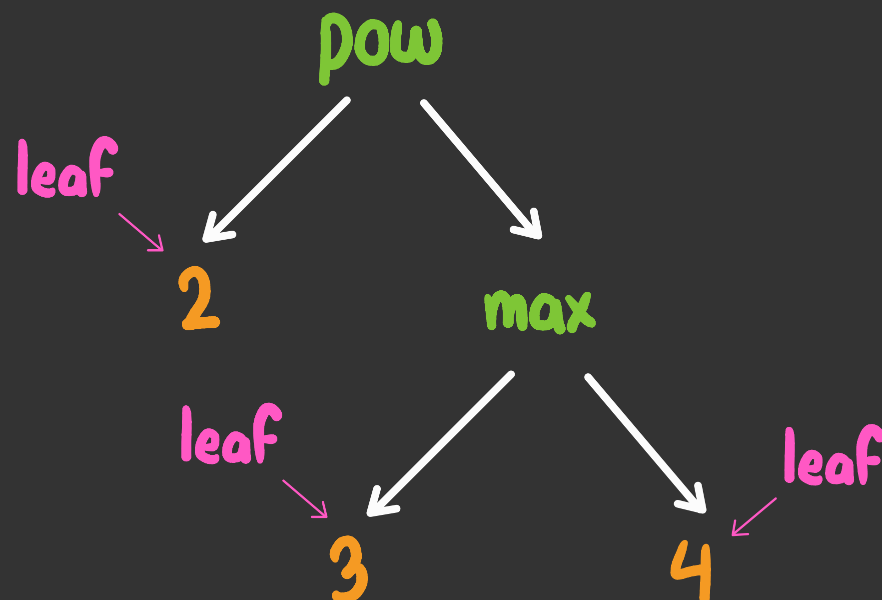


sidebar:

how to talk about trees

leaves

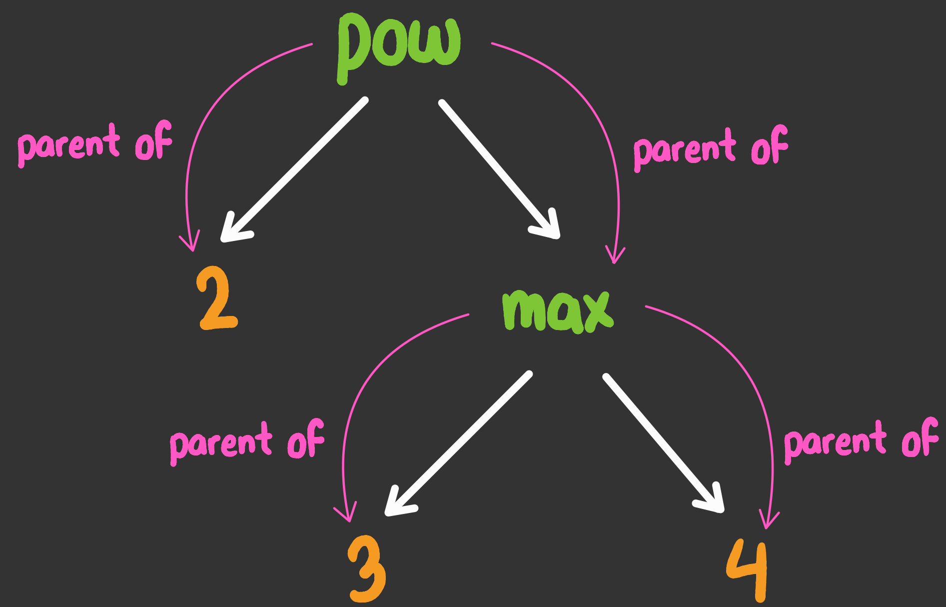
(nodes with no children)



sidebar:

how to talk about trees

parents

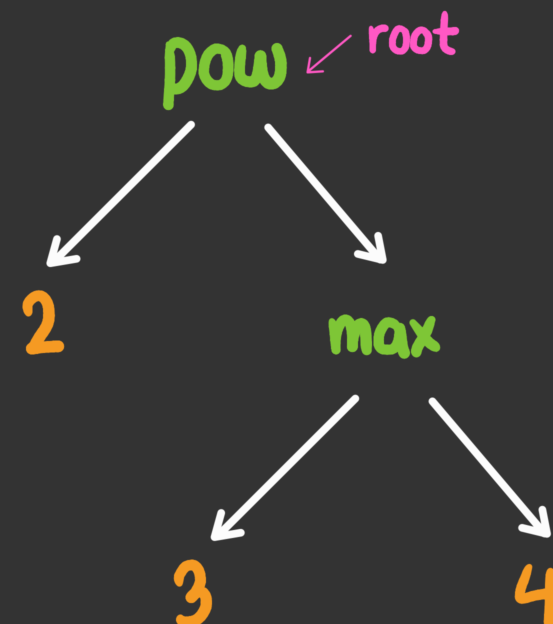


sidebar:

how to talk about trees

root

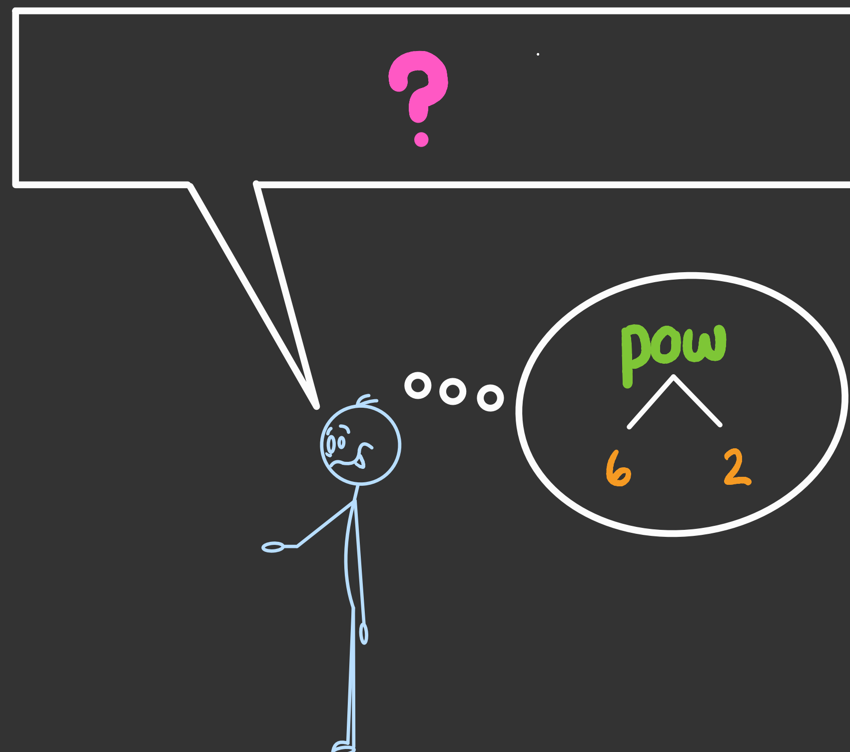
(the node with no parent)



protocol

say the label at the root,
then if the root has children:

- say beep
- communicate the first child using the protocol
- while more children remain:
  - say bop
  - communicate the next child using the protocol
- say boop



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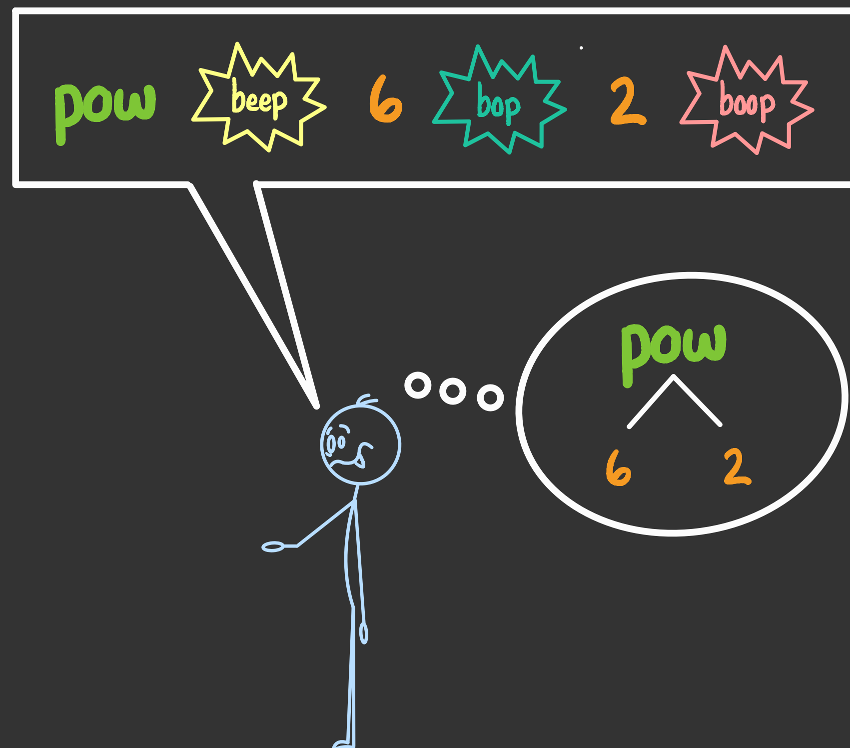

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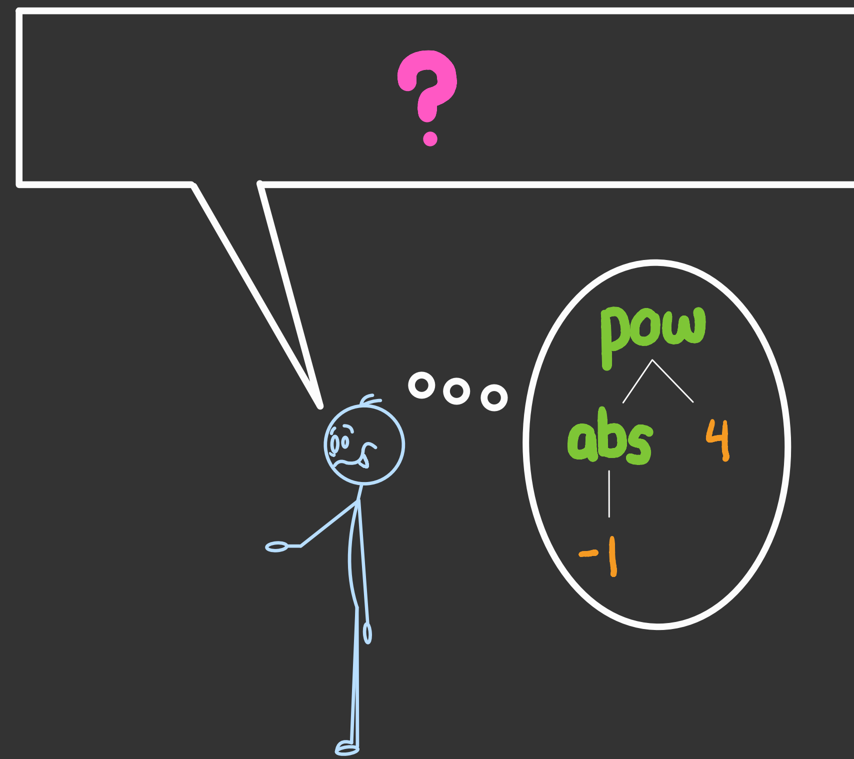

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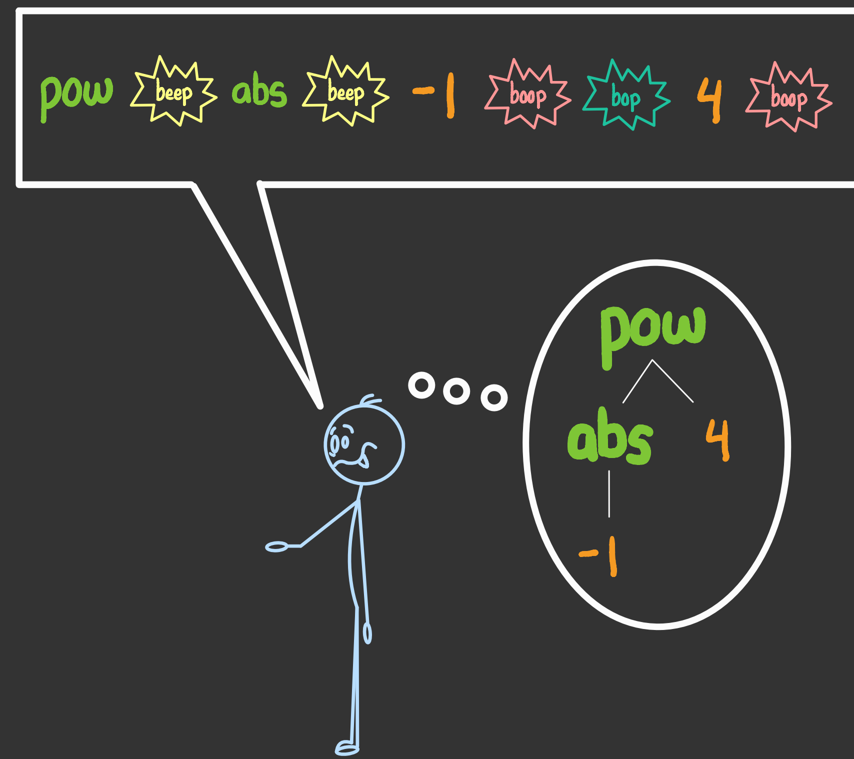

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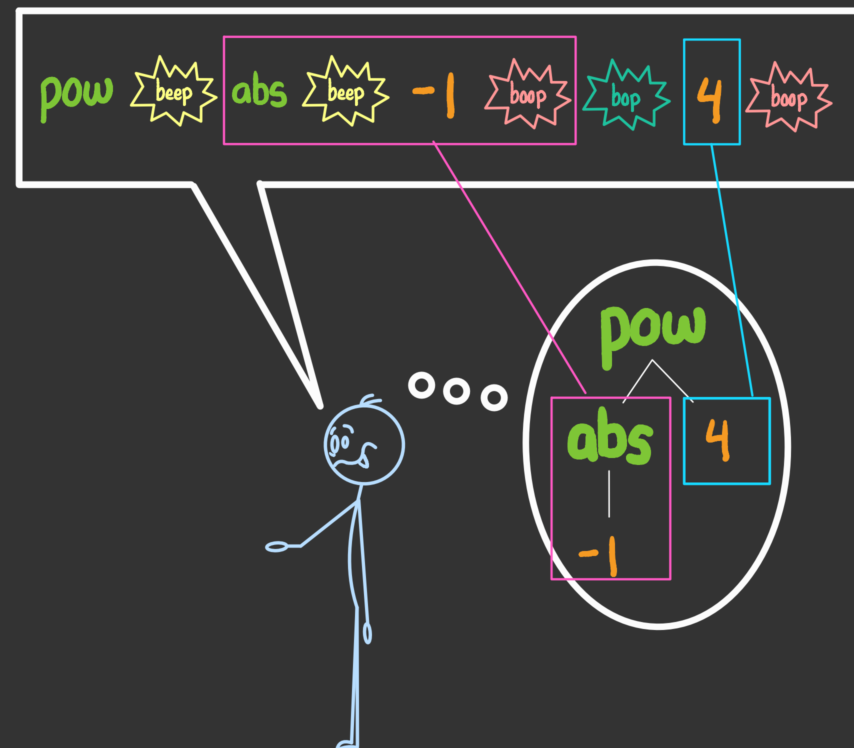
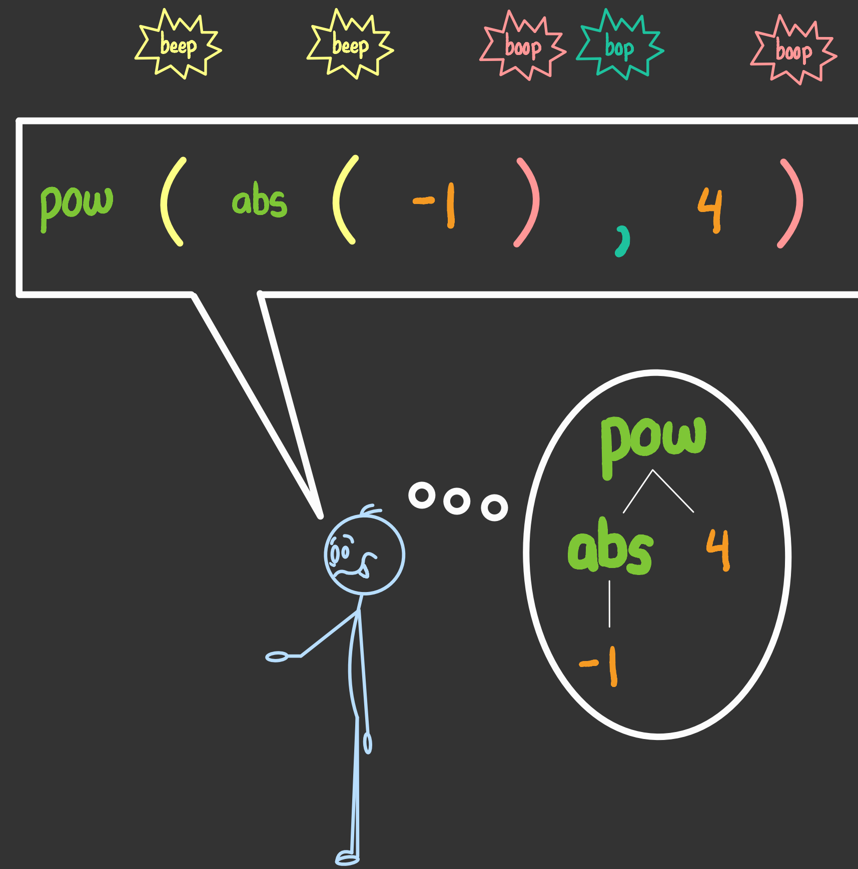


prefix protocol

type the label at the root,
then if the root has children:

- type an open paren
- communicate the first child using the protocol
- while more children remain:
    - type a comma
    - communicate the next child using the protocol
- type a closing paren



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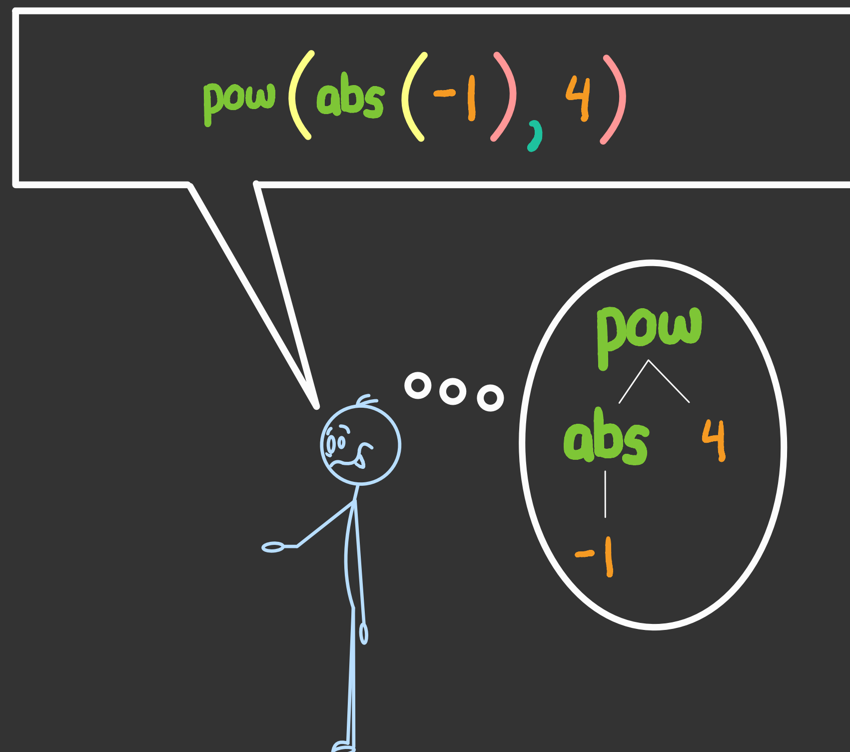

this protocol allows unambiguous communication of expression trees